

Project Design Phase-II Data Flow Diagram & User Stories

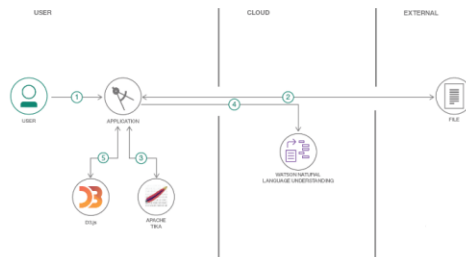
Date	25 June 2026
Team ID	SWTID-2026-8141
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

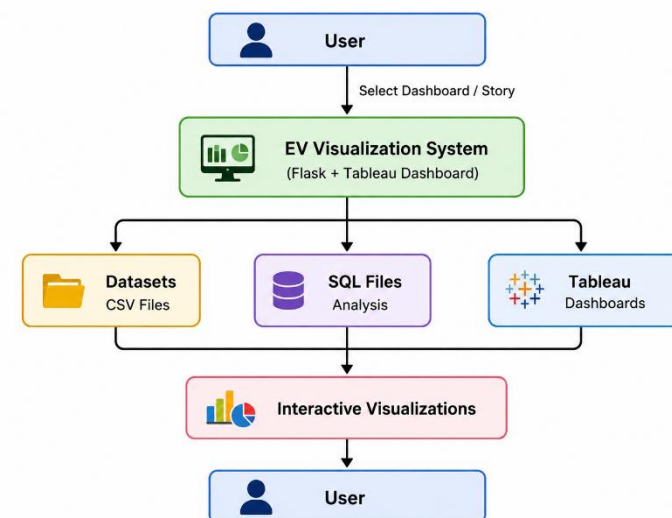
Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD level 0(Industry standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
EV Buyer	View Dashboard	USN-1	As a user, I can view the interactive EV dashboard.	Dashboard loads successfully.	High	Sprint-1
EV Buyer	Compare EV Cars	USN-2	As a user, I can compare electric vehicle range, price, and charging speed.	Vehicle comparison is displayed correctly.	High	Sprint-1
Researcher	Analyze Charging Stations	USN-3	As a researcher, I can view charging stations by region and type.	Interactive map displays charging station information.	High	Sprint-1
Data Analyst	Analyze EV Brands	USN-4	As a data analyst, I can analyze EV brands based on efficiency and number of models.	Brand analysis charts are displayed.	Medium	Sprint-2
Customer	View Price Analysis	USN-5	As a customer, I can explore electric vehicle prices across different brands.	Price charts load successfully.	High	Sprint-1
Customer	Analyze Vehicle Range	USN-6	As a customer, I can compare driving range of different EV models.	Range comparison chart is displayed.	High	Sprint-1
Policy Maker	Infrastructure Analysis	USN-7	As a policymaker, I can study charging infrastructure distribution across India.	Charging station map is interactive.	Medium	Sprint-2
Administrator	Maintain Dataset	USN-8	As an administrator, I can update the EV datasets for future analysis.	Updated data reflects in dashboards.	Low	Sprint-3